

Question 1 a)

Determine the smallest positive integer that has exactly 15 distinct positive divisors, including itself and 1.

— use tau function

$$\tau(q)\tau(r) = \tau(qr) \text{ for } q, r \text{ coprime}$$

and

$$\tau(p_1^{k_1} p_2^{k_2} \dots p_n^{k_n}) = (k_1 + 1)(k_2 + 1) \dots (k_n + 1)$$

— Prop 2.3 Ch 2
Ch 9 pg 6

$\therefore$

$\tau(n) = 15$ must have prime decomposition $p_1^{k_1} p_2^{k_2}$ where

$$\tau(p_1^{k_1} p_2^{k_2}) = (k_1 + 1)(k_2 + 1) = 15 = 3 \times 5$$

$$(k_1 + 1)(k_2 + 1) = 15$$

$$k_1 + 1 = 3 \Rightarrow k_1 = 2$$

$$k_2 + 1 = 5 \Rightarrow k_2 = 4$$

$$\Rightarrow n = 3^2 2^4 = \boxed{144}$$

or

$$k_1 + 1 = 15 \Rightarrow k_1 = 14$$

$$k_2 + 1 = 1 \Rightarrow k_2 = 0$$

$$\Rightarrow n = 2^{14} = \boxed{16,384}$$

Don't put these values

instead.

Since

$$3^2 = 9 < 16 = 2^4$$

$$3^2 2^4 < 2^8 < 2^{14}$$